

ADVANCED PROPULSION

PROFESSIONAL ELECTIVE – V

VIII Semester :										
Course Code		Category		Hours / Week		Credits	Maximum Marks			
A6AE56		PCC		L	T	P	C	CIE	SEE	Total
				3	0	0	3	40	60	100
UNIT-I		CRYOGENIC ENGINEERING								
Basic concepts. Thermo physical and fluid dynamic properties of liquid and gas hydrogen, Thermo physical and fluid dynamic properties of liquid and gas helium, Liquefaction systems of hydrogen and helium gases, Refrigeration and liquefaction principals; Joule Thomson effect and inversion curve										
UNIT-II		CRYOGENIC FLUIDS								
Properties of Cryogenic fluids, Solids at cryogenic temperatures; Superconductivity, Recuperative - Linde - Hampson, Claude, Cascade, Heylandt, Kapitza, Collins, Simon; Regenerative - Stirling cycle and refrigerator, Slova refrigerator, Gifford-McMahon refrigerator, Pulse Tube refrigerator										
UNIT-III		NUCLEAR PROPULSION SYSTEMS								
Basic concepts, components of nuclear propulsion, Fundamentals of Fission Reactions,Fission Reactor Design, fuel melting temperatures, difference between nuclear and chemical rockets, design and performance characteristics of engine										
UNIT-IV		ELECTRIC PROPULSION								
Electric rocket propulsion–Electrostatic, Electro thermal, Electromagnetic thrusters – Ion propulsion - Nuclear propulsion techniques-Solar Electric rocket propulsion–Electrostatic, Electro thermal, Electromagnetic, Neutrino Propulsion										
UNIT-V		MICRO PROPULSION AND PROPELLANT LESS PROPULSION								
Introduction ,Chemical propulsion,Solid Micro thrusters,Micro Mono-propellant Thruster, Tethers,Momentum Exchange Tether,Electrodynamics Tether										
Text Books:										
1. Martin Tajmar,Advanced Space Propulsion Systems, springer publishing 2003 2.Sutton, G. P. and Biblarj, O. Rocket propulsion elements, 7th Ed., New York: Wiley IntesciencePublications, 2001. .										
Reference Books:										
1.Mechanics & Thermodynamics of Propulsion springer publishing 2003 2.Micci, MM, Ketsdever, AD (2000) Micropropulsion for Small Spacecraft (Progress in Astronautics and Aeronautics, Vol. 187). American Institute for Aeronautics and Astronautics										
COURSE OUTCOMES : STUDENTS SHOULD BE ABLE TO:										
1. Explain the properties of helium, hydrogen at various temperatures and their liquification process. 2. Describe the properties of cryogenic fluids, Different refrigeration techniques 3. understand the concepts involved nuclear reactions and its application in space propulsion systems. 4. Illustrate the working of various electrical propulsion concepts and their application in space programs 5. Explain the concepts and working of micro propulsion svstems.										